Weak form. Find $U = (u_{\mathrm{O_2}},\, u_{\mathrm{H_2O_2}},\, \mu_{\mathrm{H}},\, \phi)$ such that

$$F(U;W) \;=\; \sum_i \mathcal{R}_i(U;v_i) \;+\; \mathcal{R}_\phi(U;w) \;=\; 0 \qquad \text{for all } W = (v_{\mathrm{O_2}}, v_{\mathrm{H_2O_2}}, v_{\mathrm{H}}, w).$$

Species i residual (Nernst–Planck):

$$\mathcal{R}_i = \underbrace{\int_\Omega \frac{c_i - c_i^{n-1}}{\Delta t}\, v_i \,\mathrm{d}x}_{\text{transient}} + \underbrace{\int_\Omega D_i\, c_i\, \nabla\big(\mu_i + \mu_i^{\mathrm{st}}\big)\cdot\nabla v_i \,\mathrm{d}x}_{\text{diffusion + electromigration + steric}} - \underbrace{\int_{\Gamma_{\mathrm{el}}} \sum_j s_{ij}\, R_j\, v_i \,\mathrm{d}s}_{\text{Butler–Volmer surface reaction}}$$

Potential residual (Poisson):

$$\mathcal{R}_\phi = \underbrace{\int_\Omega \varepsilon\, \nabla\phi\cdot\nabla w \,\mathrm{d}x}_{\text{electrostatic field}} - \underbrace{\int_\Omega \rho(\mathbf{c},\phi)\, w \,\mathrm{d}x}_{\text{space charge}} - \underbrace{\int_{\Gamma_{\mathrm{el}}} C_S\,(V_{\mathrm{app}} - \phi)\, w \,\mathrm{d}s}_{\text{Stern double layer (Robin)}}$$

Closures (the labeled pieces):

$$\mu_i = \underbrace{\ln c_i}_{\text{ideal}} + \underbrace{\tfrac{z_i F}{RT}\,\phi}_{\text{migration}} \qquad\qquad \text{proton carried as primary } \mu_{\mathrm{H}} = u_{\mathrm{H}} + \tfrac{z_{\mathrm{H}} F}{RT}\phi$$

$$\mu_i^{\mathrm{st}} = -\ln\theta, \qquad\qquad \theta = 1 - \sum_j a_j c_j - \sum_{\text{counterions}} a_b\, c_b(\phi) \quad \text{(Bikerman finite size)}$$

$$R_j = k_j^0\left[c_{\mathrm{cat}}\, e^{-\alpha_j n_j \eta_j} \prod_k \left(\frac{c_k}{c_k^{\mathrm{ref}}}\right)^{p_{jk}} - c_{\mathrm{an}}\, e^{(1-\alpha_j) n_j \eta_j}\right] \qquad\qquad \eta_j = \frac{V_{\mathrm{app}} - \phi - E_j^\circ}{V_T}, \quad |\eta_j| \le 100$$

$$\rho = F\left[\sum_i z_i c_i + \sum_{\text{counterions}} z_b\, c_b(\phi)\right] \qquad\qquad c_i = e^{u_i}, \quad c_{\mathrm{H}} = e^{\mu_{\mathrm{H}} - \frac{z_{\mathrm{H}} F}{RT}\phi}$$

Bulk Dirichlet: $c_i = c_i^\infty$, $\phi = 0$ on Γ_{bulk} (imposed strongly).

Assembled nondimensionally; $c_b(\phi)$ is the analytic Bikerman–Boltzmann counterion closure (enters the charge density and the packing fraction θ with no extra degrees of freedom). Optional water self-ionization and cation hydrolysis enter as additional volume / surface residual terms (off in the baseline).